

1 Dataset Statistics:

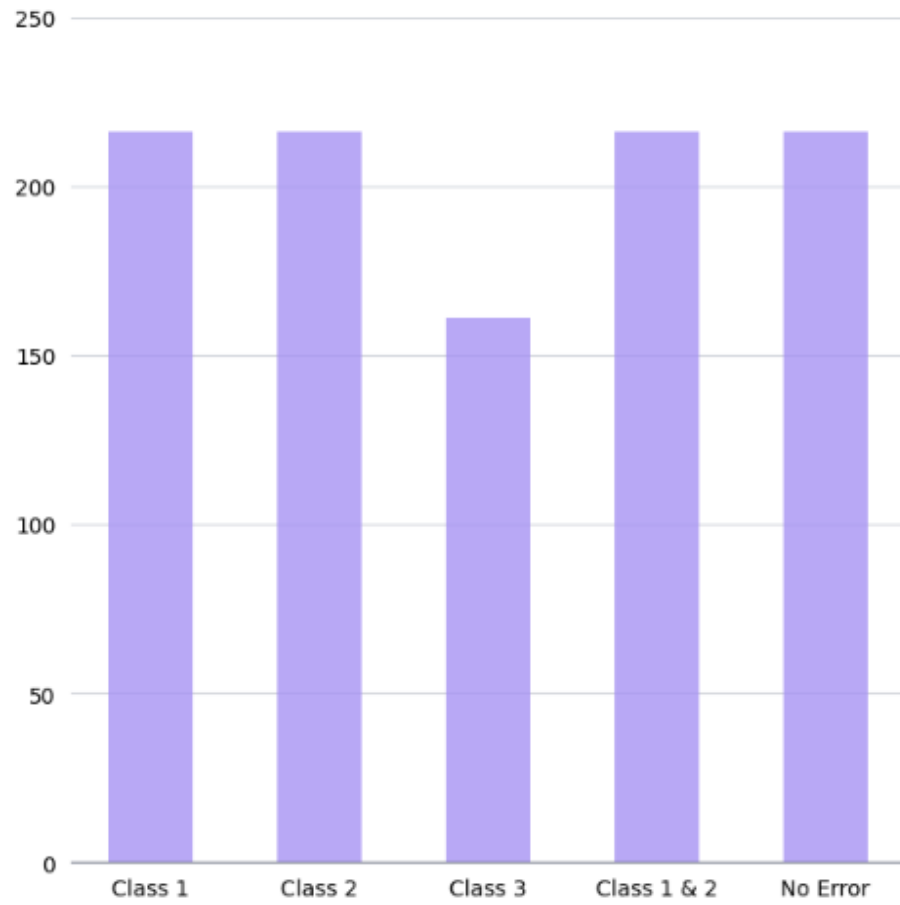


Figure 1: Class distribution of the Synthesized Error Dataset: Class 1 represents Missing Fixation, Class 2 represents Reduced Fixation Duration, and Class 3 represents Incomplete Knowledge.

2 Response time comparison

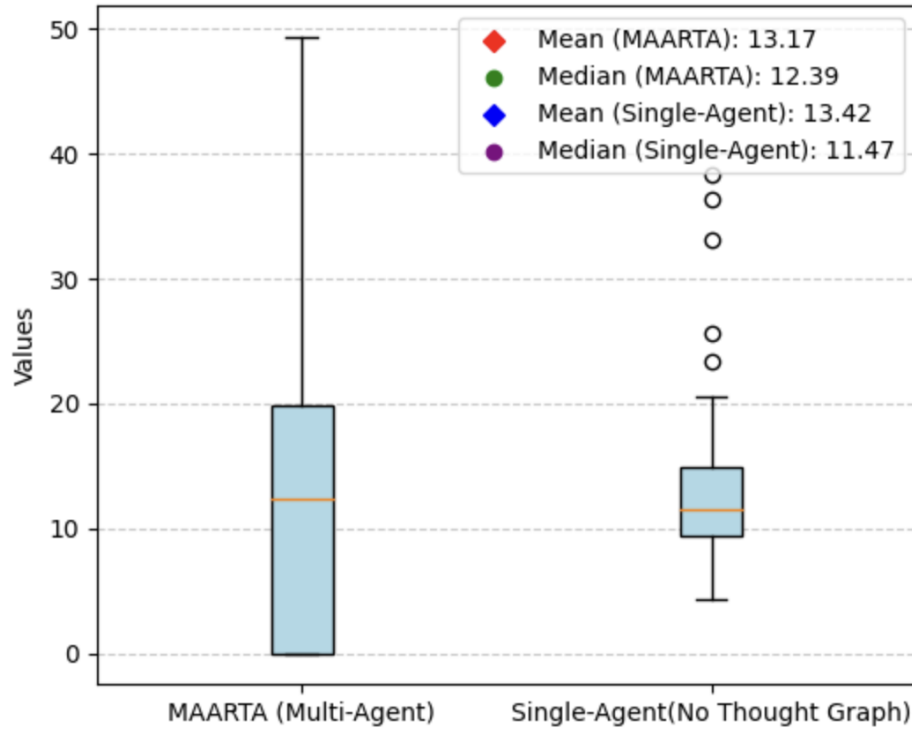


Figure 2: Response time comparison for the GPT-4o-mini model: MAARTA vs. Single-Agent GPT model with No Thought Graph used

3 Class-specific Performance:

3.1 Single Agent

Classification Report:				
	precision	recall	f1-score	
Missed diagnosis due to missing fixation	0.5565	0.7343	0.6332	
Missed diagnosis due to reduced fixation duration	0.7059	0.0837	0.1497	
Missed diagnosis due to incomplete knowledge	0.1207	0.0435	0.0639	
No missing diagnosis	0.3408	0.9167	0.4969	
micro avg	0.4427	0.4498	0.4462	
macro avg	0.4310	0.4445	0.3359	
weighted avg	0.5140	0.4498	0.3670	
samples avg	0.4454	0.4413	0.4318	
Accuracy Score: 0.30332681017612523				
Hamming Loss: 0.337573385518591				

Figure 3: Class specific performance of Mistral-7B-Instruct-v0.3.

Classification Report:				
	precision	recall	f1-score	
Missed diagnosis due to missing fixation	0.5483	0.9583	0.6976	
Missed diagnosis due to reduced fixation duration	0.6842	0.5417	0.6047	
Missed diagnosis due to incomplete knowledge	0.1852	0.7640	0.2982	
No missing diagnosis	0.9863	1.0000	0.9931	
micro avg	0.4985	0.7953	0.6129	
macro avg	0.6010	0.8160	0.6484	
weighted avg	0.6248	0.7953	0.6648	
samples avg	0.5351	0.7727	0.6125	
Accuracy Score: 0.2546341463414634				
Hamming Loss: 0.3041463414634146				

Figure 4: Class specific performance of GPT-4o-Mini.

Classification Report:				
	precision	recall	f1-score	
Missed diagnosis due to missing fixation	0.5601	0.8310	0.6692	
Missed diagnosis due to reduced fixation duration	0.6353	0.2500	0.3588	
Missed diagnosis due to incomplete knowledge	0.3889	0.0870	0.1421	
No missing diagnosis	0.6224	0.8472	0.7176	
micro avg	0.5819	0.5351	0.5575	
macro avg	0.5517	0.5038	0.4719	
weighted avg	0.5749	0.5351	0.5012	
samples avg	0.5615	0.5176	0.5249	
Accuracy Score: 0.40780487804878046				
Hamming Loss: 0.2570731707317073				

Figure 5: Class specific performance of LLAMA-3.2-11B-Vision-Instruct.

3.2 Single Agent with Thought Graph

Classification Report:				
	precision	recall	f1-score	
Missed diagnosis due to missing fixation	0.5728	0.3343	0.4222	
Missed diagnosis due to reduced fixation duration	0.5052	0.6759	0.5782	
Missed diagnosis due to incomplete knowledge	0.1414	0.3957	0.2083	
No missing diagnosis	0.3089	0.3576	0.3315	
micro avg	0.3751	0.4676	0.4163	
macro avg	0.3821	0.4409	0.3850	
weighted avg	0.4471	0.4676	0.4336	
samples avg	0.3687	0.4598	0.3873	
Accuracy Score: 0.15006002400960383				
Hamming Loss: 0.40066026410564226				

Figure 6: Class specific performance of Mistral-7B-Instruct-v0.3 with Thought Graph.

Classification Report:				
	precision	recall	f1-score	
Missed diagnosis due to missing fixation	0.5357	0.9605	0.6878	
Missed diagnosis due to reduced fixation duration	0.8740	0.5388	0.6667	
Missed diagnosis due to incomplete knowledge	0.1917	0.7848	0.3081	
No missing diagnosis	0.9014	0.8889	0.8951	
micro avg	0.5061	0.7795	0.6137	
macro avg	0.6257	0.7932	0.6394	
weighted avg	0.6727	0.7795	0.6681	
samples avg	0.5396	0.7840	0.6192	
Accuracy Score: 0.2357001972386588				
Hamming Loss: 0.2973372781065089				

Figure 7: Class specific performance of GPT-4o-Mini with Thought Graph.

Classification Report:				
	precision	recall	f1-score	
Missed diagnosis due to missing fixation	0.4638	0.9029	0.6129	
Missed diagnosis due to reduced fixation duration	0.5479	0.7079	0.6177	
Missed diagnosis due to incomplete knowledge	0.1699	0.6933	0.2730	
No missing diagnosis	0.8293	0.3333	0.4755	
micro avg	0.4113	0.7094	0.5207	
macro avg	0.5027	0.6594	0.4948	
weighted avg	0.5189	0.7094	0.5470	
samples avg	0.4311	0.6911	0.5039	
Accuracy Score: 0.16322314049586778				
Hamming Loss: 0.39462809917355374				

Figure 8: Class specific performance of LLAMA-3.2-11B-Vision-Instruct with Thought Graph.

3.3 MAARTA (Ours)

Classification Report:				
		precision	recall	f1-score
Missing	Missing Subgraph due to Missing fixation	0.74	0.42	0.54
	Subgraph due to reduced fixation duration	0.62	0.56	0.59
	Missing Subgraph due to undefined reason	0.23	0.61	0.34
	No Missing Subgraph	0.51	0.91	0.66
	micro avg	0.50	0.58	0.54
	macro avg	0.53	0.63	0.53
	weighted avg	0.59	0.58	0.55
	samples avg	0.49	0.57	0.51
Accuracy Score: 0.32974559686888455				
Hamming Loss: 0.30430528375733856				

Figure 9: Class specific performance of Mistral-7B-Instruct-v0.3 with MAARTA.

Classification Report:				
		precision	recall	f1-score
Missing Subgraph	Missing Subgraph due to Missing fixation	0.70	0.95	0.81
	Missing Subgraph due to reduced fixation duration	0.88	0.96	0.92
	Missing Subgraph due to undefined reason	0.77	0.57	0.66
	No Missing Subgraph	0.91	0.99	0.95
	micro avg	0.80	0.91	0.85
	macro avg	0.82	0.87	0.83
	weighted avg	0.81	0.91	0.85
	samples avg	0.83	0.90	0.85
Accuracy Score: 0.7473170731707317				
Hamming Loss: 0.09463414634146342				

Figure 10: Class specific performance of GPT-4o-Mini with MAARTA.

Classification Report:				
	precision	recall	f1-score	
Missing Subgraph due to Missing fixation	0.69	0.87	0.77	
Missing Subgraph due to reduced fixation duration	0.65	0.83	0.72	
Missing Subgraph due to undefined reason	0.37	0.49	0.42	
No Missing Subgraph	0.77	0.96	0.86	
micro avg	0.64	0.82	0.72	
macro avg	0.62	0.79	0.69	
weighted avg	0.65	0.82	0.72	
samples avg	0.68	0.82	0.72	
Accuracy Score: 0.5024679170779862				
Hamming Loss: 0.1920039486673248				

Figure 11: Class specific performance of LLAMA-3.2-11B-Vision-Instruct with MAARTA.